FedPASA: Reliability-Aware Semantic Adaptation for Cross-Domain Federated Recommendation

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Abstract

Federated recommendation (FR) enables collaborative model training while keeping user interactions local. However, transferring recommendation knowledge across services remains challenging due to disjoint item catalogs and uneven target-domain supervision. Frequently observed items provide abundant interaction evidence, whereas low-frequency items offer limited supervision, and client-specific preferences further introduce heterogeneity into semantic adaptation. Existing approaches commonly employ shared transformations or client-level personalization without explicitly accounting for differences in item-level supervision. To address these challenges, we propose FedPASA, a reliability-aware semantic adaptation framework for cross-domain federated recommendation. FedPASA introduces Popularity-Gated Dual-path Routing (PGDR), which combines two residual adaptation paths using item-frequency-derived weights. A private client profile further conditions the adapted representations through feature-wise scaling and shifting, while AutoAnchor incorporates pretrained semantic information through catalog-level anchoring. Extensive experiments on real-world cross-domain recommendation benchmarks demonstrate the effectiveness of FedPASA. The proposed framework achieves superior average cross-domain recommendation performance over competitive baselines under both sampled and full-catalog ranking. Further analyses demonstrate that frequency-aware routing improves recommendation accuracy beyond additional adapter capacity, while client conditioning and semantic anchoring provide complementary adaptation capabilities.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Distributed computing methodologies.*

Keywords

Federated Recommendation; Cross-Domain Recommendation; Semantic Transfer; Long-tail Recommendation

ACM Reference Format:

Anonymous Author(s). 2027. FedPASA: Reliability-Aware Semantic Adaptation for Cross-Domain Federated Recommendation. In *Proceedings of the ACM Web Conference 2027 (WWW '27).* ACM, New York, NY, USA, 12 pages.

1 Introduction

Recommender systems learn user–item preferences from interaction histories [10] and support online services. Centralized training, however, raises concerns about sharing these histories. Federated recommendation (FR) keeps interactions local and aggregates model parameters [1, 24]. Recent work develops client co-clustering [11], distributed heterogeneous-graph learning [33], and communication-efficient low-rank updates [26], advancing collaborative recommendation without pooling raw histories.

Cross-domain transfer is another longstanding need: platforms wish to reuse a mature catalog to bootstrap a new service with limited behavioral evidence. Many cross-domain methods transfer preferences through overlapping users or aligned representations [3, 23, 34]. Such links may be unavailable across services, while federated constraints prevent pooling user histories. We study disjoint item catalogs without requiring linked users between source and target domains.

Text-based transferable representations offer an alternative to identifier alignment [12, 14, 38]. However, semantic representations alone do not resolve uneven target supervision. Interaction imbalance and conformity can distort representation learning [34, 35]: frequent items supply many ranking constraints, whereas rare items provide little adaptation evidence. Clients also observe different semantic regions, so a globally rare item may be familiar to one client yet unseen by another. Excessive adaptation can move sparse items away from useful pretrained semantics. Adaptation should therefore account for item-level evidence, private client context, and the amount of semantic prior to preserve.

Hierarchical personalization, personalized adapters, and expert mixtures address client variation [19, 32, 36]. Client-level personalization alone does not specify how items with different training frequencies should share adaptation paths: a shared transformation uses the same parameterization for all items, while adding an adapter increases capacity without determining its allocation. We instead use item frequency as an observable proxy for target supervision, route items between two residual paths, and condition the routed representation on the local interaction profile.

To address the challenges mentioned above, we propose FedPASA, which combines three operations in a shared semantic adapter (Figure 1). Specifically, Popularity-Gated Dual-path Routing (PGDR) mixes a source-initialized path and a target-initialized path using a frequency-derived weight; client conditioning produces feature-wise scales and shifts from a locally pooled semantic profile; in addition, AutoAnchor interpolates the resulting representation with the frozen text embedding, with the anchor strength set by a catalog-level user-to-item ratio. Both the source-initialized and target-initialized residual paths are jointly optimized using target-domain interactions.

We evaluate this design on multiple real-world datasets, with controls that separate frequency assignment, model capacity, and final personalization. Our contributions are summarized as follows:

- We propose FedPASA, a reliability-aware semantic adaptation framework that adjusts item representations according to evidence strength at the item, client, and catalog levels.

WWW '27, Dublin, Ireland
2027.

The framework addresses uneven target supervision: frequent and sparse items contribute different amounts of evidence, while local interaction histories determine the semantic context in which that evidence is observed. The proposed framework applies to different backbone encoders.

- We design PGDR to combine two trainable residual adaptation paths using frequency-derived weights, without an additional learned routing network. Client conditioning adjusts the combined representations using local semantic profiles, while AutoAnchor controls their mixture with the frozen semantic prior through a catalog-level rule.
- We conduct extensive experiments on multiple real-world datasets. With the same text encoder, FedPASA achieves higher average cross-domain recommendation accuracy than the evaluated baselines. Comparisons that retain the same network structure and parameter count further show that frequency-derived mixing weights outperform using a common pair of weights for all items or shuffling the item–weight assignments. Ablation studies further characterize the effects of individual components.

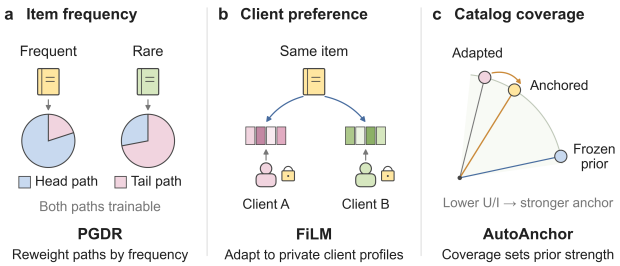


Figure 1: Three adaptation controls: (a) frequency-based path mixing; (b) private-profile conditioning; (c) catalog-dependent prior preservation.

2 Related Work

2.1 Federated Semantic Transfer

Federated recommendation (FR) enables collaborative learning from decentralized user interactions while preserving data privacy [1, 24]. Existing studies address client heterogeneity through proximal optimization and personalized local models [2, 5, 15, 16]. Cross-domain FR further explores knowledge transfer across heterogeneous item catalogs. TransFR combines pretrained language representations, adapter tuning, and local personalization [38]. FedDCSR disentangles shared and domain-specific sequential factors [39], while PFCR and FFMSR exploit textual content for semantic transfer [7, 21]. FedCRF enhances cross-domain knowledge fusion through semantic graphs and contrastive learning [8]. FeDecider further introduces personalized aggregation of directional LoRA updates to address domain heterogeneity [9].

Despite these advances, existing methods primarily focus on semantic representation learning, knowledge alignment, and personalized aggregation, while item-level supervision differences remain less explored. Frequent and infrequent items provide different amounts of interaction evidence, making uniform semantic adaptation insufficient for capturing their distinct learning requirements. To address this limitation, FedPASA introduces popularity-gated dual-path routing (PGDR), which allocates source-initialized and target-initialized adaptation paths according to item frequency.

2.2 Personalization and Preservation

Personalized FR aims to capture heterogeneous user preferences through client-specific adaptation and knowledge sharing. FedPA combines private and group adapters with adaptive gating [36], while FedVLR employs personalized multimodal expert fusion [19]. Other studies explore long-tail recommendation through ESAM [4], user preference adaptation through MeLU [13], and expert specialization through multi-gate mixture-of-experts architectures [22]. However, these approaches mainly address personalization or item imbalance separately, without explicitly coordinating item-frequency-dependent adaptation with private client preferences. FedPASA combines PGDR with client conditioning, generating feature-wise scales and shifts from local semantic profiles based on FiLM [28].

Preserving transferred semantic knowledge is another important aspect of cross-domain adaptation. L2-SP and DELTA retain source information through parameter and feature regularization [17, 18]. However, these methods constrain parameter updates or intermediate representations without directly controlling the contribution of frozen semantic features to final ranking representations. To address this distinction, FedPASA introduces AutoAnchor, which adjusts the semantic prior contribution through catalog-dependent interpolation according to target-domain coverage.

3 FedPASA

3.1 Problem Setup and Overview

We consider cross-domain federated recommendation with disjoint item catalogs. Let C, $\mathcal{U}$, and $\mathcal{I}$ denote the sets of clients, users, and target items, respectively. Each client c maintains a subset of users $\mathcal{U}_c$ and their interaction histories locally. A frozen text encoder maps each item i to a unit-normalized semantic representation $x_i \in \mathbb{R}^d$, providing a common representation space across domains. Each user u is associated with a private embedding $e_u \in \mathbb{R}^d$ that remains on its corresponding client.

FedPASA adapts the frozen semantic representations through three mechanisms: Popularity-Gated Dual-path Routing (PGDR), Private-Profile Client Conditioning (FiLM), and Catalog-Level Semantic Anchoring (AutoAnchor). Source-domain training initializes one residual adaptation path, while the second path is randomly initialized on the target domain. Both paths remain trainable during target-domain federated optimization.

For user u on client c, the recommendation score is defined as

$$s(u,i) = e_u^\top z_{c,i}^{\text{anchor}}. \tag{1}$$

The model is optimized using Bayesian Personalized Ranking (BPR) [30]:

$$\mathcal{L}_{\text{BPR}} = -\mathbb{E}_{(u,i,j)} \log \sigma\big(\tau[s(u,i) - s(u,j)]\big),$$

where i is an observed positive item, j a sampled negative item, and τ controls the score scale.

Figure 2 illustrates the overall FedPASA framework, and Table 5 summarizes the notation. The forward computation consists of

three consecutive stages:

$$x_i \xrightarrow{\text{routing}} h_i \xrightarrow[\text{and residual}]{\text{FiLM}} z_{c,i} \xrightarrow{\text{anchoring}} z_{c,i}^{\text{anchor}}. \tag{2}$$

3.2 Popularity-Gated Dual-path Routing

Target items exhibit substantially different interaction frequencies, resulting in different amounts of supervision for semantic adaptation. PGDR explicitly uses item frequency to determine the relative contributions of two residual adaptation paths, enabling frequency-dependent representation learning within a shared model.

Let $\text{freq}(i)$ denote the number of target-domain training interactions containing item i. We first construct an inverse-frequency score $\bar{q}_i = 1/\sqrt{\text{freq}(i)+1}$ and normalize it over the target catalog:

$$q_i = \frac{\bar{q}_i - \min_j \bar{q}_j}{\max_j \bar{q}_j - \min_j \bar{q}_j + \epsilon}. \tag{3}$$

A threshold t and exponent κ further reshape the routing score:

$$q_i' = \text{clip}\big(\tfrac{q_i - t}{1-t}, 0, 1\big)^{\kappa}.$$

The resulting score q_i' increases as item frequency decreases and controls the relative contribution of the target-initialized path.

PGDR contains two residual adapters, denoted by A_h and A_l. The head adapter A_h is initialized from the source-domain adapter, providing transferred semantic adaptation knowledge. The tail adapter A_l is randomly initialized on the target domain and learns complementary transformations from target interactions. Each adapter contains two sequential residual blocks, $v \leftarrow v + F_\ell(v)$, where F_ℓ is a bottleneck multilayer perceptron, followed by ℓ_2 normalization. Both adapters are optimized during target-domain federated training.

The routed item representation is

$$h_i = (1 - q_i')\, A_h(x_i) + q_i'\, A_l(x_i). \tag{4}$$

Frequently observed items place greater weight on the source-initialized head path, whereas lower-frequency items receive a larger contribution from the target-initialized tail path. The mixed representation h_i is subsequently passed to the client-conditioning module.

3.3 Private-Profile Client Conditioning

Item frequency characterizes supervision at the item level, while clients may observe substantially different semantic regions of the target catalog. To incorporate this client-side heterogeneity, FedPASA constructs a private semantic profile from each client's local interaction history and uses it to condition the routed representation.

For client c, let $\mathcal{I}_c$ denote the multiset of items appearing in its training interactions. Repeated interactions retain their multiplicity in the profile construction. The client profile is defined as

$$p_c = \text{norm}\Big(\tfrac{1}{|\mathcal{I}_c|} \sum_{i \in \mathcal{I}_c} x_i\Big). \tag{5}$$

A shared bottleneck network $g_\phi : \mathbb{R}^d \to \mathbb{R}^{2d}$ maps the profile into feature-wise FiLM parameters:

$$(\gamma_c, \beta_c) = g_\phi(p_c). \tag{6}$$

The routed representation h_i is then modulated as

$$\begin{aligned} m_{c,i} &= h_i \odot (1 + \lambda \tanh \gamma_c) + \lambda \beta_c, \\ z_{c,i} &= \text{norm}\big((1-\alpha) A_h(x_i) + \alpha m_{c,i}\big). \end{aligned} \tag{7}$$

The coefficient λ controls the strength of client-dependent modulation. The tanh transformation bounds the multiplicative factor to $[1-\lambda, 1+\lambda]$, while the additive term provides learned client-specific feature shifts. The coefficient $\alpha \in [0,1]$ controls the contribution of the conditioned representation relative to the head-path residual.

For the conditioning module, only the shared parameters ϕ participate in federated communication. The client profile p_c is computed from local training interactions and remains on the client.

3.4 Catalog-Level Semantic Anchoring

After frequency-aware routing and client conditioning, FedPASA further integrates the frozen semantic representation into the final ranking representation. AutoAnchor determines the contribution of this semantic prior according to target-catalog coverage.

We characterize target-catalog coverage using the user-to-item ratio

$$\eta_T = |\mathcal{U}_T| / |\mathcal{I}_T|,$$

where $|\mathcal{U}_T|$ denotes the number of observed target-domain users and $|\mathcal{I}_T|$ denotes the target catalog size. Given coverage thresholds $\tau_l < \tau_h$ and maximum anchor weight ρ, AutoAnchor computes

$$\begin{aligned} \rho_{\text{eff}} &= \rho \cdot \text{clip}\Big(\frac{\tau_h - \eta_T}{\tau_h - \tau_l}, 0, 1\Big), \\ z_{c,i}^{\text{anchor}} &= \text{norm}\big((1 - \rho_{\text{eff}})\, z_{c,i} + \rho_{\text{eff}}\, x_i\big). \end{aligned} \tag{8}$$

The effective weight ρ_{eff} controls the contribution of the frozen semantic prior. When $\eta_T \le \tau_l$, the maximum anchor weight ρ is used. As η_T increases between τ_l and τ_h, the prior contribution decreases linearly. For $\eta_T \ge \tau_h$, the final representation relies directly on the adapted representation. Its weight is determined from target-domain catalog statistics and remains fixed throughout federated training and evaluation.

The default catalog-level gate introduces no additional trainable parameters. We additionally evaluate a per-item learnable alternative,

$$\rho_i = \rho\, \sigma(a_i),$$

where $a \in \mathbb{R}^{|\mathcal{I}|}$ contains item-specific logits included in federated aggregation. Table 24 compares catalog-level and per-item anchoring.

3.5 Federated Training and Local Personalization

FedPASA shares $\theta = \{\theta_h, \theta_l, \phi\}$, comprising the two residual adapters and client-conditioning network. Each client c maintains a private user-embedding table $E_c = \{e_u : u \in \mathcal{U}_c\}$.

At round r, clients initialize θ_c from the broadcast parameters $\theta^{(r)}$, retain E_c from the preceding round, and take K BPR steps updating both. Only θ_c is uploaded. Under full participation, the server applies equal-client FedAvg:

$$\theta^{(r+1)} = \frac{1}{|C|} \sum_{c \in C} \theta_c.$$

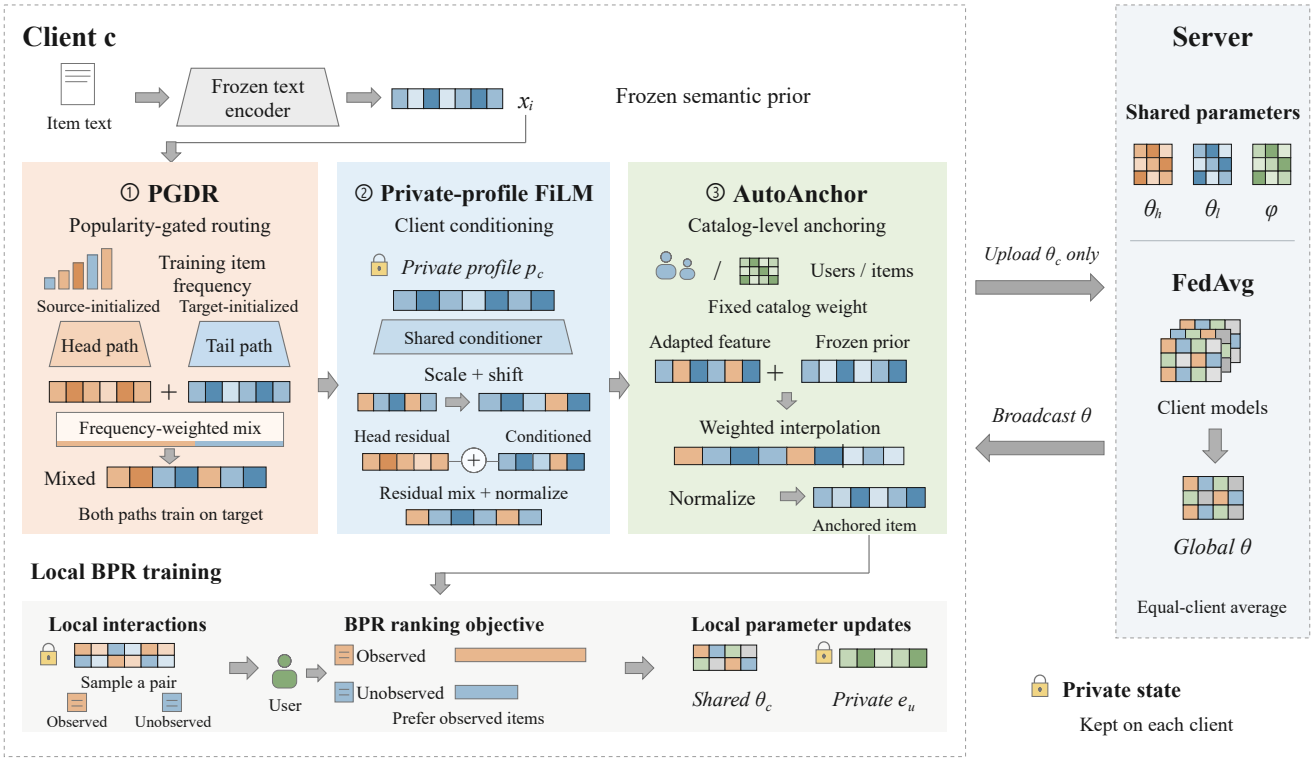


Figure 2: FedPASA framework. PGDR combines source-initialized and target-initialized residual adapters according to item frequency. Private-profile FiLM conditions the resulting representations on local interaction semantics, while AutoAnchor integrates the frozen semantic prior. Clients optimize shared model parameters and private user embeddings using BPR, and the server aggregates the shared parameters through FedAvg.

The federated adaptation stage optimizes the BPR objective without an additional proximal penalty.

After R communication rounds, the resulting global parameters $\theta^{(R)}$ define the shared FedPASA model. We refer to this model before final local personalization as *FedPASA base*. For final personalization, each client initializes its local model from $\theta^{(R)}$, retains its learned private user embeddings, and optimizes

$$\mathcal{L}_{\text{pers},c} = \mathcal{L}_{\text{BPR},c} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2.$$

The proximal term constrains θ_c around the final global solution [15, 16]. Each client takes K_p steps updating both θ_c and E_c; the resulting personalized parameters remain local.

Algorithm 1 summarizes the complete training procedure. Concrete optimization settings are provided in the experimental setup (§4.2).

Initialization and local optimization. Source-domain training uses one residual adapter together with source-domain user embeddings. The resulting adapter parameters initialize the target-domain head path. The tail path is randomly initialized. The final linear layer of the conditioning network is zero-initialized so that client modulation starts from the shared representation.

Target item frequencies are computed exclusively from target training interactions. Each client profile is computed from its own training-item multiset, with repeated interactions retained. Routing statistics, client profiles, and the catalog-level anchor weight are fixed throughout federated optimization, personalization, and evaluation.

Private user embeddings persist across communication rounds. AdamW optimizer states are reset at the beginning of each local optimization phase. During local training, users are sampled uniformly within a client and positive items are sampled uniformly from the corresponding user's training history. Negative items are sampled from items outside the user's observed training interactions. Dropout is enabled during model updates and disabled during evaluation.

The communication process exchanges only the shared adaptation and conditioning parameters. Raw interaction histories, client profiles, and private user embeddings remain on their respective clients.

3.6 Analysis of the Adaptation Controls

We analyze the main adaptation mechanisms from three perspectives: the relationship between target-estimation reliability and semantic-prior weighting, the geometry of normalized semantic anchoring, and the interaction between popularity routing and client conditioning.

Algorithm 1 FedPASA training and local personalization

Require: Frozen item features, training histories, clients C, rounds R, local steps K, personalization steps K_p, proximal coefficient μ_p

For cross-domain training, optimize a source-domain residual adapter by source BPR and equal-client FedAvg
Initialize θ_h from the source adapter; randomly initialize θ_l and ϕ
Compute target routing scores q_i from target training frequencies
Compute each client profile p_c from its training interactions
Compute and fix the catalog-level anchor weight ρ_{eff}
Initialize private user tables $E_c = \{e_u : u \in \mathcal{U}_c\}$ for all clients
$\theta^{(0)} \leftarrow \{\theta_h, \theta_l, \phi\}$
for $r = 0, \ldots, R-1$ **do**
  Broadcast $\theta^{(r)}$ to all participating clients
  for all $c \in C$ **in parallel do**
    $\theta_c \leftarrow \theta^{(r)}$; retain E_c from the preceding round
    Take K local steps on $\mathcal{L}_{\text{BPR}}$, updating (θ_c, E_c)
    Keep E_c local and upload θ_c
  end for
  $\theta^{(r+1)} \leftarrow |C|^{-1} \sum_{c \in C} \theta_c$
end for
Preserve $\theta^{(R)}$ as the FedPASA base model
for all $c \in C$ **do**
  $\theta_c \leftarrow \theta^{(R)}$; retain the final E_c
  $\ell_c \leftarrow \mathcal{L}_{\text{BPR}} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2$
  Take K_p local personalization steps, updating (θ_c, E_c)
end for
return Personalized (θ_c, E_c) stored on each client

3.6.1 *Bias–Variance Tradeoff of Prior Weighting.*

Proposition 1 (Bias–variance tradeoff). *Let z^* denote an ideal target representation and x a semantic prior. Consider an estimated target representation $\hat{z} = z^* + \varepsilon$, where $\mathbb{E}\varepsilon = 0$ and $\mathbb{E}\|\varepsilon\|_2^2 = \sigma^2/r$ with $r, \sigma^2 > 0$. Here, r characterizes the reliability of the target estimate: larger r corresponds to lower estimation variance. Let $b^2 = \|x - z^*\|_2^2$ denote the squared bias of the semantic prior. For the interpolated estimate $\tilde{z}_\omega = (1-\omega)\hat{z} + \omega x$, we have*

$$\mathbb{E}\|\tilde{z}_\omega - z^*\|_2^2 = (1-\omega)^2\sigma^2/r + \omega^2 b^2, \qquad \omega^* = \frac{\sigma^2}{\sigma^2 + rb^2}. \tag{9}$$

When $b > 0$, ω^* decreases as r increases.

The optimal weight increases with estimation noise and decreases with prior bias: when target evidence is limited, more of a useful prior should be retained. AutoAnchor operationalizes this principle at the catalog level by using target-catalog coverage to determine the contribution of the frozen semantic prior.

3.6.2 *Geometry of Normalized Semantic Anchoring.*

Proposition 2 (Normalized anchor geometry). *For fixed unit vectors x and z, define $w_\omega = (1-\omega)z + \omega x$, $y_\omega = w_\omega/\|w_\omega\|_2$, and $a = x^\top z$. For $0 \le \omega < 1/2$,*

$$\|w_\omega\|_2 \ge 1 - 2\omega > 0, \qquad \frac{d(x^\top y_\omega)}{d\omega} = \frac{(1-\omega)(1-a^2)}{\|w_\omega\|_2^3} \ge 0. \tag{10}$$

Therefore, $1 - x^\top y_\omega$ is non-increasing with respect to the anchor weight ω.

This proposition characterizes the geometric effect of normalized semantic interpolation: for a fixed adapted representation, the cosine distance to the frozen prior is non-increasing with the anchor weight. The experimental analysis (§4.4.2) further examines the representation geometry produced after federated optimization.

Table 1: Dataset statistics after filtering, including training, validation, and test interactions.

Dataset	# Users	# Items	# Interactions
Music	12,315	10,680	339,923
Books	262,698	170,389	10,848,388
Movies	28,226	19,252	1,056,608

3.6.3 *Client Conditioning.*

The multiplicative part of the FiLM transformation in Eq. (7) is $1 + \lambda \tanh \gamma_c$. Since $\tanh \gamma_c \in [-1, 1]$, the resulting feature-wise multiplicative factors satisfy $1 - \lambda \le 1 + \lambda \tanh \gamma_c \le 1 + \lambda$. Thus, λ directly controls the range of client-dependent multiplicative modulation, while the learned additive term $\lambda\beta_c$ shifts the routed representation according to the client's semantic profile.

3.6.4 *Interaction of Routing and Client Conditioning.*

PGDR and client conditioning act on the same semantic representation but control different aspects of adaptation. To make the interaction explicit, consider the case in which FiLM modulation is disabled, i.e., $\lambda = 0$. Combining Eqs. (4) and (7) gives

$$z_{c,i} = \text{norm}\big((1 - \alpha q_i')A_h(x_i) + \alpha q_i' A_l(x_i)\big). \tag{11}$$

The effective coefficient of the target-initialized path is therefore $\alpha q_i'$. The item-frequency-derived score q_i' determines how the two adaptation paths are allocated to each item, while α controls the overall strength of this allocation. Even for the largest routing score $q_i' = 1$, the head path retains a contribution of $1 - \alpha$.

When FiLM is enabled, client-dependent feature scales and shifts are applied to the frequency-routed representation before normalization and semantic anchoring. Consequently, item frequency controls path combination, whereas the private client profile controls feature-level adjustment.

4 Experiments

We evaluate FedPASA through four research questions:

- **RQ1:** How does FedPASA compare with federated recommendation and semantic-transfer baselines?
- **RQ2:** How much does each component contribute beyond additional adapter capacity?
- **RQ3:** How do key hyperparameters affect the recommendation performance of FedPASA?
- **RQ4:** How does FedPASA retain private client information during collaborative learning?

4.1 Datasets

We evaluate cross-domain recommendation on three Amazon datasets. The benchmark contains all six ordered transfers among Amazon Music (CDs & Vinyl), Books, and Movies (Movies & TV) [27]; their item catalogs are disjoint and no linked users are required. Table 1 summarizes the retained data. Appendix B provides additional preprocessing and evaluation details.

4.2 Experimental Settings

4.2.1 *Implementation Details.*

We treat the retained review interactions as implicit feedback and exclude users with fewer than

Table 2: Overall performance comparison on three cross-domain transfer tasks. The best and second-best results are highlighted in bold and underline, respectively.

Transfer	Music→Books		Music→Movies		Books→Movies	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [9]	0.2011	0.0714	<u>0.4278</u>	<u>0.1823</u>	<u>0.3981</u>	<u>0.1705</u>
FedMF [1]	0.2005	0.0715	0.2127	0.0792	0.2127	0.0792
FedAvg [24]	0.1949	0.0672	0.3440	0.1388	0.3440	0.1388
FedProx [16]	0.1998	0.0705	0.3381	0.1351	0.3381	0.1351
FedAdam [29]	0.1986	0.0690	0.3398	0.1359	0.3398	0.1359
TransFR [38]	<u>0.2058</u>	<u>0.0723</u>	0.3292	0.1283	0.3330	0.1307
L2-SP [17]	0.2032	0.0718	0.3244	0.1244	0.3141	0.1229
FedPer [2]	0.2031	0.0714	0.3410	0.1353	0.3492	0.1397
pFedMe [5]	0.2047	0.0719	0.3284	0.1285	0.3368	0.1332
FedMoE [25]	0.2039	0.0707	0.3424	0.1355	0.3566	0.1444
FedPASA	**0.2719**	**0.1159**	**0.4980**	**0.2254**	**0.4948**	**0.2284**

Table 3: Overall performance comparison on the remaining three cross-domain transfer tasks and the average results across all six transfer directions.

Transfer	Books→Music		Movies→Books		Movies→Music		Six-transfer mean	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [9]	<u>0.5355</u>	<u>0.2283</u>	0.2058	<u>0.0734</u>	<u>0.5380</u>	<u>0.2284</u>	<u>0.3844</u>	<u>0.1590</u>
FedMF [1]	0.2212	0.0826	0.2005	0.0715	0.2212	0.0826	0.2115	0.0778
FedAvg [24]	0.5074	0.2129	0.1949	0.0672	0.5074	0.2129	0.3488	0.1396
FedProx [16]	0.5024	0.2093	0.1998	0.0705	0.5024	0.2093	0.3468	0.1383
FedAdam [29]	0.5073	0.2098	0.1986	0.0690	0.5073	0.2098	0.3485	0.1382
TransFR [38]	0.5046	0.2121	0.2001	0.0705	0.5048	0.2115	0.3462	0.1376
L2-SP [17]	0.4996	0.2083	0.2017	0.0710	0.5040	0.2122	0.3412	0.1351
FedPer [2]	0.5052	0.2150	0.2040	0.0721	0.5027	0.2099	0.3509	0.1406
pFedMe [5]	0.5054	0.2131	0.2026	0.0712	0.5064	0.2117	0.3474	0.1383
FedMoE [25]	0.5088	0.2143	<u>0.2064</u>	0.0731	0.5107	0.2141	0.3548	0.1420
FedPASA	**0.5408**	**0.2403**	**0.2824**	**0.1249**	**0.5462**	**0.2439**	**0.4390**	**0.1965**

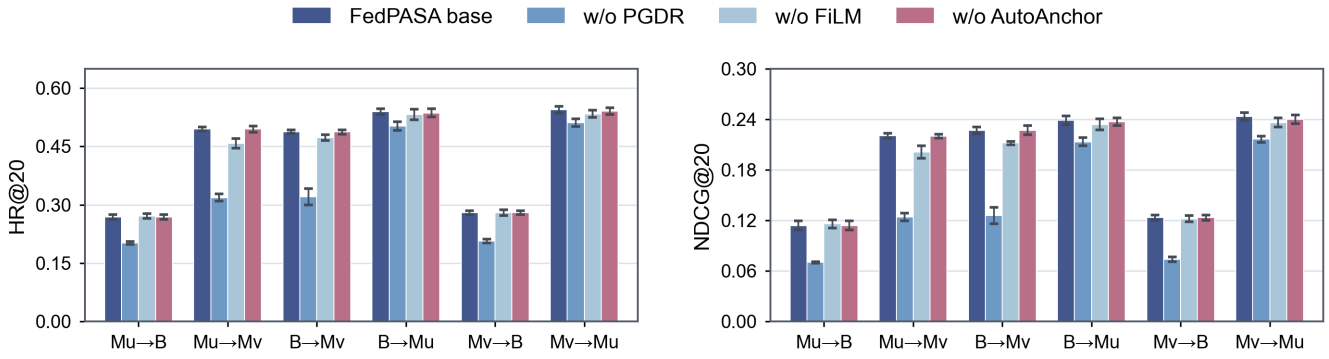


Figure 3: Ablation study across six cross-domain transfer tasks. Error bars indicate standard deviations over five runs. Mu, B, and Mv denote Music, Books, and Movies, respectively.

three interactions in each domain. Interaction histories are sorted chronologically, with the latest interaction held out for testing, the preceding interaction for validation, and the remaining interactions used for training. We evaluate HR@20 and NDCG@20 under both sampled and full-catalog ranking. Sampled ranking pairs each test item with 99 randomly sampled negative items, excluding the user's training and validation items.

4.2.2 Baselines. We compare FedPASA with representative recent and classical federated recommendation baselines. **FeDecider** [9]

Table 4: Frequency–exposure ablation under full-catalog ranking before personalization. C denotes low-frequency items (≤ 5) observed locally, U denotes low-frequency items (≤ 5) unobserved locally, and W denotes higher-frequency items (> 5).

Model	C: low-frequency / seen		U: low-frequency / unseen		W: frequency > 5	
	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FedPASA base	**0.0160**	**0.0070**	**0.0347**	**0.0151**	**0.0102**	**0.0037**
w/o PGDR	0.0041	0.0010	0.0039	0.0013	0.0023	0.0007
w/o FiLM	0.0041	0.0018	0.0276	0.0123	0.0037	0.0014
w/o AutoAnchor	0.0082	0.0023	0.0144	0.0047	0.0039	0.0014

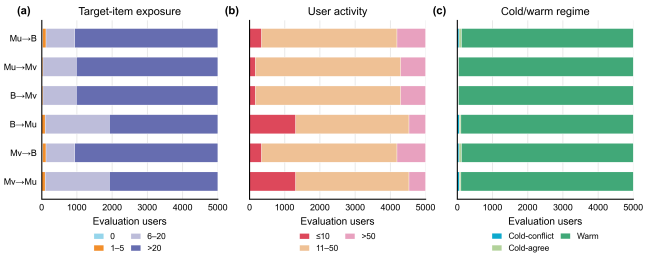


Figure 4: Evaluation-user composition by target-item frequency, user activity, and cold/warm groups. Mu/B/Mv denote Music/Books/Movies; C/U correspond to Table 4.

employs LLM representations and personalized directional LoRA aggregation for cross-domain recommendation. **TransFR** [38] transfers pretrained language-model adapters and performs local personalization. **FFMSR** [21] leverages textual semantics for federated cross-domain recommendation. **FedDCSR** [39] disentangles shared and domain-specific representations for sequential recommendation. **FedMoE** [25] employs heterogeneous mixture-of-experts models with personalized expert selection and modular aggregation. **FedAdam** [29] applies adaptive server-side optimization, while **FedProx** [16] introduces proximal regularization to address client heterogeneity. **pFedMe** [5] learns personalized federated models through Moreau-envelope optimization. **FedPer** [2] combines globally shared representations with client-specific personalization layers. **L2-SP** [17] regularizes target parameters toward their source initialization. **FedAvg** [24] aggregates local model updates to learn a shared global model, while **FedMF** [1] performs federated matrix factorization for collaborative recommendation.

4.3 Main Results and Discussion (RQ1)

Evidence-dependent adaptation improves transfer across disjoint catalogs. FedPASA achieves the highest HR and NDCG on all six sampled-ranking transfers (Tables 2 and 3), improving mean HR by 14.2% over FeDecider. The largest gains occur on Books and Movies. PGDR allocates adaptation paths according to the available item evidence, while private-profile conditioning incorporates each client's semantic context. The gains in HR and NDCG indicate improved retrieval of relevant items and their placement near the top of the ranked list.

The advantage extends beyond sampled candidates. FedPASA also achieves the highest HR and NDCG in the full-catalog comparison (Tables 6–8). Here, each held-out item competes with the full set of eligible catalog items. The consistent improvements under sampled and full-catalog evaluation demonstrate that FedPASA maintains its ranking advantage across different candidate-set sizes. The shared semantic space supports ranking across disjoint catalogs, while target interactions guide evidence-dependent adaptation.

4.4 Ablation Study (RQ2)

Figure 3 evaluates the base model and its component ablations before local personalization, retaining the BPR objective and remaining modules. Tables 2–3 evaluate the final personalized models.
(1) w/o PGDR. Bypassing the tail path makes routing independent of item frequency:

$$q'_i = 0, \qquad h_i = A_h(x_i). \tag{12}$$

Both metrics decrease on every transfer, consistent with losing item-dependent path allocation. PGDR also exceeds constant and shuffled routing (Table 16), which retain both paths but remove item dependence or break the frequency–weight correspondence. These controls link the gain to frequency assignment.
(2) w/o FiLM. Setting $\lambda = 0$ gives $m_{c,i} = h_i$ and

$$z_{c,i} = \mathrm{norm}\big((1 - \alpha q'_i)A_h(x_i) + \alpha q'_i A_l(x_i)\big). \tag{13}$$

Each item now shares one representation across clients. Restoring FiLM improves accuracy on Movies and Music by introducing profile-specific feature scales and shifts. This modulation complements item-level routing with the semantic context of local interactions.
(3) w/o AutoAnchor. Removing the prior mixture yields

$$\rho_{\text{eff}} = 0, \qquad z_{c,i}^{\text{anchor}} = z_{c,i}. \tag{14}$$

This variant ranks items using the adapted representation. The complete model achieves higher six-transfer mean HR and NDCG (Table 16). AutoAnchor combines the learned target features with pretrained semantic information, using catalog coverage to set their relative contributions.

4.4.1 Evidence Mechanism Analysis. Table 4 retains historical records organized by global frequency and local familiarity. Component effects under a common evaluation protocol are assessed in Figure 3 and Table 16.
Evaluation composition motivates group-wise analysis. Figure 4 summarizes the evaluation-user distribution across six transfer directions. Most evaluation users have frequent target items (> 20 training occurrences), moderate activity (11–50 training interactions), and warm targets (frequency > 5). Transfers to Music contain larger proportions of users with target-item frequencies of 6–20 and low activity (≤ 10 interactions) than transfers to Books or Movies. The

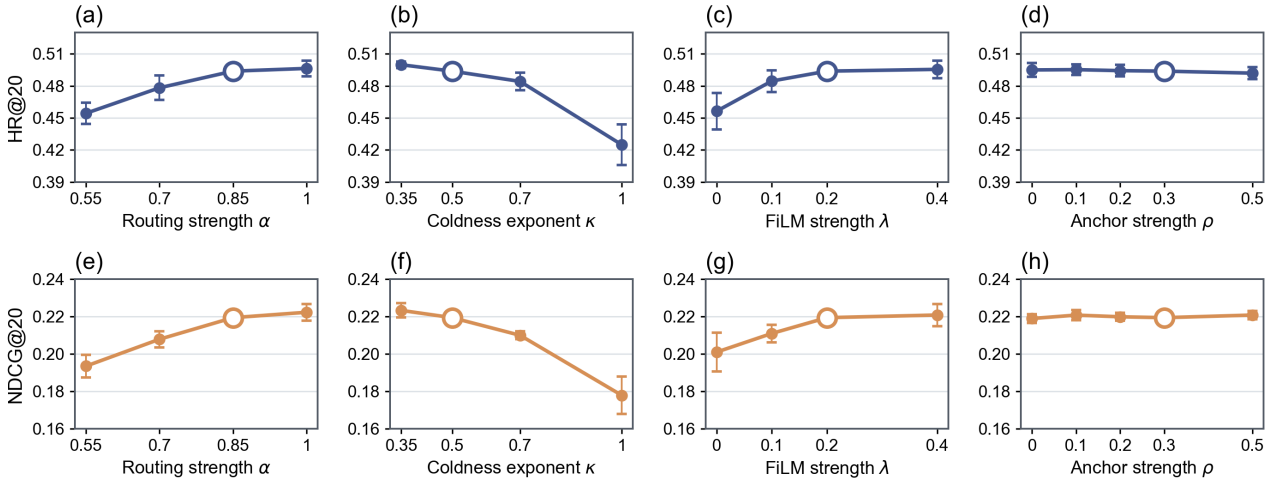


Figure 5: Hyperparameter sensitivity of FedPASA on Music→Movies. The upper and lower panels show HR@20 and NDCG@20, respectively. Error bars indicate standard deviations over three runs, and open circles mark the validation-selected values.

predominance of warm targets motivates reporting group-wise performance alongside aggregate metrics (Table 4).

4.4.2 AutoAnchor and representation drift. We measure representation drift through cosine distance from the frozen prior:

$$D = \frac{1}{|C||\mathcal{I}|} \sum_{c \in C} \sum_{i \in \mathcal{I}} \left(1 - \cos(z_{c,i}^{\text{anchor}}, x_i)\right). \tag{15}$$

Table 25 reports a smaller cold-item distance and a larger all-item distance for FedPASA base than for the no-anchor variant after training. Proposition 2 characterizes the interpolation operation itself: increasing the anchor weight yields non-decreasing alignment with the prior for fixed adapted features.

4.5 Hyperparameter Sensitivity

Figure 5 evaluates routing strength α, coldness exponent κ, client-conditioning strength λ, and anchoring strength ρ on Music→Movies.

Increasing α improves both metrics, with diminishing gains at higher values, supporting greater use of the routed, client-conditioned representation. Larger κ instead reduces accuracy: it suppresses intermediate routing scores and shifts weight toward the source-initialized path, consistent with insufficient target-specific adaptation.

Client conditioning improves performance before leveling off around $\lambda = 0.2$, suggesting that moderate profile-dependent modulation captures most of the benefit. Performance remains stable across the tested anchoring strengths, indicating lower sensitivity to ρ on this transfer. These trends favor prioritizing routing parameters during tuning.

4.6 Privacy Preserving Analysis (RQ4)

In the proposed FedPASA, privacy is supported by separating shared adaptation parameters from private client information. Specifically, after completing local training, each client uploads the parameters of the two residual adapters and the shared FiLM network. The server aggregates these parameters through FedAvg and broadcasts a single global model for the next round. Raw interaction histories, client profiles, and user embeddings remain on their respective clients, enabling collaborative representation learning without pooling users' behavioral records.

For personalized adaptation, each client computes its semantic profile and FiLM modulation locally. Final personalization further updates the model and user embeddings without another upload, retaining client-specific representations on the client.

For stronger protection, update normalization can be performed locally before upload. Moreover, FedPASA is compatible with local differential privacy [6]: clients can clip the concatenated adapter and FiLM update to a bounded norm and apply an ε_r-LDP randomizer calibrated to this bound. Under this extension, the server receives privatized updates rather than raw local updates. Allocating per-round budgets such that $\sum_{r=1}^{R} \varepsilon_r \leq \varepsilon$ provides an explicit ε-LDP guarantee for each client's communication transcript across training rounds.

5 Conclusion

We propose FedPASA for adapting semantic representations across disjoint catalogs under uneven target supervision. PGDR allocates two trainable residual paths by item frequency, private-profile conditioning captures client preferences, and AutoAnchor controls the frozen prior's contribution through catalog coverage. Together, these operations incorporate item evidence and local context while keeping interaction histories private. Experiments across six transfer directions show improved average accuracy under sampled and full-catalog ranking. Equal-capacity controls identify frequency-dependent routing as the main source of improvement; component and group analyses characterize the contributions of conditioning and anchoring. Sensitivity results further highlight routing allocation as a key control, while moderate client conditioning captures most gains and anchoring shows lower sensitivity within the tested range. FedPASA thus offers a practical framework for evidence-aware federated semantic transfer. Future work will extend this design to evolving catalogs and dynamic client profiles.

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A Proofs and Mechanism Analysis

We use the definitions and assumptions in §3.6.

A.1 Proof of Proposition 1

Let $v = \sigma^2/r > 0$. The zero-mean error eliminates the cross term:

$$R(\omega) = (1-\omega)^2 v + \omega^2 b^2. \tag{16}$$

Thus $R'(\omega) = 2[(v+b^2)\omega - v]$ and $R''(\omega) = 2(v+b^2) > 0$. The unique minimizer on $[0,1]$ is $\omega^* = v/(v+b^2)$, including $\omega^* = 1$ when $b = 0$. Moreover,

$$\frac{d\omega^*}{dr} = -\frac{\sigma^2 b^2}{(\sigma^2 + rb^2)^2} \le 0. \tag{17}$$

The derivative is negative for $b > 0$.

A.2 Proof of Proposition 2

Let $d_\omega = \|w_\omega\|$ and $n_\omega = (1-\omega)a + \omega$. The reverse triangle inequality gives $d_\omega \ge 1 - 2\omega > 0$, and

$$d_\omega^2 = (1-\omega)^2 + \omega^2 + 2\omega(1-\omega)a.$$

Using $n'_\omega = 1 - a$ and $d'_\omega = (2\omega - 1)(1-a)/d_\omega$, the quotient rule gives

$$\frac{d(x^\top y_\omega)}{d\omega} = \frac{(1-\omega)(1-a^2)}{d_\omega^3} \ge 0. \tag{18}$$

Since $|a| \le 1$, alignment is non-decreasing for fixed x, z.

A.3 Routing Gradients at Fixed Capacity

With FiLM disabled, Eq. (11) gives $v_i = (1-a_i)A_h(x_i) + a_i A_l(x_i)$, where $a_i = \alpha q'_i$ is fixed during training. For $g_i = \nabla_{v_i}\ell$, the chain rule gives

$$\begin{aligned} \nabla_{\theta_h}\ell &= (1-a_i) J_h(x_i)^\top g_i, \\ \nabla_{\theta_l}\ell &= a_i J_l(x_i)^\top g_i, \end{aligned} \tag{19}$$

where J_h, J_l are parameter Jacobians. Constant routing removes item-dependent gradient allocation; shuffling preserves capacity and the gate distribution but changes their association with item evidence.

A.4 Bounds for Client Conditioning

Write $s_c = \tanh \gamma_c$ and $m_c(h) = h + \lambda(s_c \odot h + \beta_c)$. Because h mixes unit vectors, $\|h\|_2 \le 1$. Using $\|s_c\|_\infty \le 1$ and $\lambda \ge 0$ gives

$$\begin{aligned} \|m_c(h) - h\|_2 &\le \lambda\big(\|s_c \odot h\|_2 + \|\beta_c\|_2\big) \\ &\le \lambda(1 + \|\beta_c\|_2). \end{aligned} \tag{20}$$

For fixed client parameters, the additive shift cancels:

$$\|m_c(h) - m_c(\tilde{h})\|_2 \le (1+\lambda)\|h - \tilde{h}\|_2. \tag{21}$$

Thus λ controls pre-normalization sensitivity and client-specific shifts.

B Dataset Statistics

Table 1 reports retained users after the three-interaction filter; their last two events form validation and test sets. AutoAnchor divides this user count by catalog size.

Ranking masks training and validation items (§4.2). DistilBERT uses normalized masked-mean features with at most 96 tokens; LLaMA-2-7B features use a fixed signed block projection to 512 dimensions.

Metric aggregation. For held-out rank r_u, HR@K averages $\mathbf{1}[r_u \le K]$ and NDCG@K averages $\mathbf{1}[r_u \le K]/\log_2(r_u+1)$. For a partition into nonempty groups $\mathcal{G}$,

$$M = \sum_{G \in \mathcal{G}} \frac{|G|}{|\mathcal{U}_{\text{eval}}|} M_G, \qquad M \in \{\text{HR@}K, \text{NDCG@}K\}.$$

Thus group size weights each group mean M_G in the aggregate.

Table 5: Notation used in FedPASA.

Symbol	Definition
x_i, e_u	Frozen unit item vector; private user vector
A_h, A_l, F_ℓ	Head/tail adapters; residual bottleneck MLP
q_i, q'_i, t, κ	Coldness scores, threshold, and exponent
$h_i, m_{c,i}, z_{c,i}$	Routed, modulated, and normalized features
p_c, g_ϕ	Private profile and shared conditioner
α, λ	Mixture and client-modulation strengths
$\eta_T, \rho, \rho_{\text{eff}}, \tau_l, \tau_h$	Coverage, anchor weights, and thresholds
θ, μ, μ_p	Shared parameters and proximal coefficients
r, b^2, σ^2	Reliability, prior bias, and noise scale

C Full Ranking and Run-to-Run Variation

Tables 6–8 report full ranking: M/B/V denote Music/Books/Movies; H/N denote HR@20/NDCG@20; bold/underline mark best/second-best results. FeDecider uses three runs; others use five. FedPASA includes personalization.

Table 6: Full ranking on M→B, M→V, B→V.

	M→B		M→V		B→V	
Method	H	N	H	N	H	N
FedMF [1]	<u>0.0003</u>	<u>0.0001</u>	0.0022	0.0009	0.0022	0.0009
FedAvg [24]	0.0002	0.0000	0.0037	0.0015	0.0037	0.0015
FedProx [16]	0.0001	0.0000	0.0041	0.0015	0.0041	0.0015
FedAdam [29]	0.0001	0.0000	0.0033	0.0011	0.0033	0.0011
TransFR [38]	0.0001	0.0000	0.0037	0.0012	0.0030	0.0011
L2-SP [17]	0.0000	0.0000	0.0029	0.0011	0.0028	0.0011
FedPer [2]	0.0001	0.0000	0.0034	0.0012	0.0040	0.0014
Prox-FT	0.0001	0.0000	0.0029	0.0009	0.0034	0.0012
MoE-2	0.0001	0.0000	0.0041	0.0015	0.0045	0.0016
FeDecider [9]	0.0001	0.0000	<u>0.0071</u>	<u>0.0025</u>	<u>0.0055</u>	<u>0.0019</u>
FedPASA	**0.0017**	**0.0006**	**0.0126**	**0.0046**	**0.0140**	**0.0050**

Table 7: Full ranking on B→M, V→B, V→M.

	B→M		V→B		V→M	
Method	H	N	H	N	H	N
FedMF [1]	0.0024	0.0009	<u>0.0003</u>	<u>0.0001</u>	0.0024	0.0009
FedAvg [24]	0.0125	0.0043	0.0002	0.0000	0.0125	0.0043
FedProx [16]	0.0130	0.0047	0.0001	0.0000	0.0130	0.0047
FedAdam [29]	0.0110	0.0041	0.0001	0.0000	0.0110	0.0041
TransFR [38]	0.0126	0.0047	0.0000	0.0000	0.0132	0.0050
L2-SP [17]	0.0132	0.0046	0.0001	0.0000	0.0132	0.0049
FedPer [2]	0.0127	0.0045	0.0002	0.0001	0.0119	0.0042
Prox-FT	0.0125	0.0046	0.0000	0.0000	0.0132	0.0049
MoE-2	0.0134	<u>0.0050</u>	0.0002	0.0001	0.0137	0.0052
FeDecider [9]	<u>0.0136</u>	0.0049	0.0001	0.0001	<u>0.0145</u>	<u>0.0055</u>
FedPASA	**0.0177**	**0.0065**	**0.0018**	**0.0009**	**0.0188**	**0.0069**

Table 8: Full-ranking means over six transfers.

Method	HR@20	NDCG@20
FedMF [1]	0.0016	0.0006
FedAvg [24]	0.0055	0.0019
FedProx [16]	0.0057	0.0021
FedAdam [29]	0.0048	0.0018
TransFR [38]	0.0054	0.0020
L2-SP [17]	0.0054	0.0019
FedPer [2]	0.0054	0.0019
Prox-FT	0.0053	0.0019
MoE-2	0.0060	0.0022
FeDecider [9]	<u>0.0068</u>	<u>0.0025</u>
FedPASA	**0.0111**	**0.0041**

Sampled-ranking variability is reported below as mean ± sample SD. For each method, the SD of the six-transfer mean is computed across run-level means.

Table 9: Sampled HR@20 variability: baselines.

Transfer	TransFR	MoE-2	FeDecider
M→B	0.2058 ± 0.0051	0.2039 ± 0.0032	0.2011 ± 0.0023
M→V	0.3292 ± 0.0065	0.3424 ± 0.0120	0.4278 ± 0.0055
B→V	0.3330 ± 0.0210	0.3566 ± 0.0077	0.3981 ± 0.0131
B→M	0.5046 ± 0.0127	0.5088 ± 0.0056	0.5355 ± 0.0240
V→B	0.2001 ± 0.0071	0.2064 ± 0.0059	0.2058 ± 0.0029
V→M	0.5048 ± 0.0074	0.5107 ± 0.0085	0.5380 ± 0.0214
Mean	0.3462 ± 0.0066	0.3548 ± 0.0042	0.3844 ± 0.0070

Table 10: Sampled HR@20 variability: FedPASA variants.

Transfer	FedPASA base	FedPASA
M→B	0.2692 ± 0.0063	0.2719 ± 0.0068
M→V	0.4950 ± 0.0055	0.4980 ± 0.0085
B→V	0.4881 ± 0.0039	0.4948 ± 0.0053
B→M	0.5394 ± 0.0072	0.5408 ± 0.0126
V→B	0.2796 ± 0.0053	0.2824 ± 0.0052
V→M	0.5449 ± 0.0085	0.5462 ± 0.0094
Mean	0.4360 ± 0.0044	0.4390 ± 0.0058

Table 11: Sampled NDCG@20 variability: baselines.

Transfer	TransFR	MoE-2	FeDecider
M→B	0.0723 ± 0.0032	0.0707 ± 0.0010	0.0714 ± 0.0014
M→V	0.1283 ± 0.0028	0.1355 ± 0.0035	0.1823 ± 0.0055
B→V	0.1307 ± 0.0090	0.1444 ± 0.0027	0.1705 ± 0.0093
B→M	0.2121 ± 0.0044	0.2143 ± 0.0031	0.2283 ± 0.0099
V→B	0.0705 ± 0.0030	0.0731 ± 0.0024	0.0734 ± 0.0027
V→M	0.2115 ± 0.0042	0.2141 ± 0.0059	0.2284 ± 0.0095
Mean	0.1376 ± 0.0027	0.1420 ± 0.0019	0.1590 ± 0.0042

Table 12: Sampled NDCG@20 variability: FedPASA variants.

Transfer	FedPASA base	FedPASA
M→B	0.1140 ± 0.0053	0.1159 ± 0.0054
M→V	0.2204 ± 0.0031	0.2254 ± 0.0053
B→V	0.2268 ± 0.0042	0.2284 ± 0.0032
B→M	0.2389 ± 0.0051	0.2403 ± 0.0076
V→B	0.1232 ± 0.0034	0.1249 ± 0.0035
V→M	0.2432 ± 0.0046	0.2439 ± 0.0069
Mean	0.1944 ± 0.0032	0.1965 ± 0.0035

Evaluation configuration. FedAvg, FedProx, FedAdam, TransFR, L2-SP, FedPer, Prox-FT, MoE-2, and FedPASA share frozen DistilBERT features, user partitions, chronological splits, evaluation users, and candidate construction. FedMF shares the data splits and evaluation setup and uses learned item factors. MoE-2 mixes two shared residual adapters with $g(x) = \mathrm{softmax}(Wx + b)$. It is an expert-mixture control, not a reproduction of FedMoE [25], whose design includes client-specific submodel search and modular aggregation. Prox-FT applies 40 post-federation proximal steps; it is a proximal fine-tuning control rather than the Moreau-envelope optimization of pFedMe. Published comparisons appear in Tables 14–15 and 17–19; sequential configurations appear in Appendix E.2.

D Additional Baseline Comparisons and Ablations

Table 13: Native sequential benchmarks: full-ranking HR@20/NDCG@20. FFMSR uses two runs; FedDCSR uses one run with target-indexed scores.

Transfer	FFMSR [21]		FedDCSR [39]	
	HR@20	NDCG@20	HR@20	NDCG@20
M→B	0.0273	0.0116	0.0254	0.0218
M→V	0.1354	0.0824	0.0924	0.0724
B→V	0.1231	0.0720	0.0924	0.0724
B→M	0.1585	0.0841	0.1090	0.0832
V→B	0.0214	0.0094	0.0254	0.0218
V→M	0.1701	0.0992	0.1090	0.0832
Mean	0.1060	0.0598	0.0756	0.0591

Table 14: Published Amazon Music/Books/Movies HR@10 [38].

Method	M→B	M→V	B→V	Mean
CMF [31]	0.1833	0.1378	0.1455	0.1555
EMCDR [23]	0.2149	0.1893	0.1652	0.1898
DisenCDR [3]	0.2243	0.1988	0.1912	0.2048
FCF [1]	0.1380	0.1154	0.1004	0.1179
IFedRec [37]	0.1981	0.1810	0.1703	0.1831
FELLRec [40]	0.2003	0.1844	0.1731	0.1859
FedCT [20]	0.2034	0.1765	0.1585	0.1795
FedDCSR [39]	0.2217	0.1881	0.1719	0.1939
TransFR [38]	**0.2335**	**0.2123**	**0.2089**	**0.2182**

Table 15: Published Amazon Music/Books/Movies NDCG@10 [38].

Method	M→B	M→V	B→V	Mean
CMF [31]	0.0955	0.0764	0.0662	0.0794
EMCDR [23]	0.1121	0.0901	0.0705	0.0909
DisenCDR [3]	**0.1182**	0.0923	0.0821	0.0975
FCF [1]	0.0663	0.0583	0.0421	0.0556
IFedRec [37]	0.0834	0.0673	0.0497	0.0668
FELLRec [40]	0.0995	0.0812	0.0831	0.0879
FedCT [20]	0.1122	0.0899	0.0845	0.0955
FedDCSR [39]	0.1085	0.0892	0.0732	0.0903
TransFR [38]	0.1133	**0.0934**	**0.0893**	**0.0987**

Initialization and adaptation. Table 16 shows similar means, slightly favoring source-free training. Both variants adapt pretrained text features on the target; these results support target-side adaptation, without establishing an additional accuracy gain from source initialization.

Table 16: Component and capacity controls: 99-negative ranking, six-transfer means over five runs. The final row includes personalization.

Model	HR@20	NDCG@20
FedPASA base	0.4360	0.1944
w/o PGDR	0.3434	0.1373
w/o FiLM	0.4245	0.1867
w/o AutoAnchor	0.4346	0.1934
Constant routing	0.3481	0.1404
Shuffled routing	0.3381	0.1350
Per-item anchor	0.4365	0.1950
Source-free base	0.4362	0.1958
MoE-2	0.3548	0.1420
FedPASA	0.4390	0.1965

Table 17: Published FFMSR and PFCR recommendation performance. R: recall; N: NDCG.

FFMSR [21]				
Domain	R10	R50	N10	N50
Office	0.1312	0.2069	0.0890	0.1054
Arts	0.1257	0.2377	0.0711	0.0955
OnlineRetail	0.1655	0.4110	0.0763	0.1300
Pantry	0.0695	0.1818	0.0326	0.0568
PFCR (full prompting) [7]				
Domain	R10	R50	N10	N50
Office	0.1206	0.1938	0.0782	0.0941
Arts	0.1153	0.2199	0.0677	0.0904
OnlineRetail	0.1561	0.3982	0.0739	0.1266
Pantry	0.0616	0.1591	0.0293	0.0503

Table 18: Published FedCRF full-ranking Recall [8].

Pair / target	Recall@10	Recall@20
Kitchen–Food: Kitchen	0.04465	0.06601
Kitchen–Food: Food	0.11078	0.16105
Care–Beauty: Care	0.07115	0.09897
Care–Beauty: Beauty	0.09342	0.13384
Food–OnlineRetail: Food	0.11142	0.16185
Food–OnlineRetail: OnlineRetail	0.22864	0.31488

Table 19: Published FedCRF full-ranking NDCG [8].

Pair / target	NDCG@10	NDCG@20
Kitchen–Food: Kitchen	0.02523	0.03112
Kitchen–Food: Food	0.06708	0.08193
Care–Beauty: Care	0.04250	0.05001
Care–Beauty: Beauty	0.05398	0.06513
Food–OnlineRetail: Food	0.06713	0.08189
Food–OnlineRetail: OnlineRetail	0.19232	0.23368

Table 20: Fine-grained frequency slices (99-negative ranking; unpersonalized models).

Frequency	Method	HR@20	NDCG@20
Unseen	TransFR [38]	0.3020	0.1070
	MoE-2	0.2460	0.0927
	FedPASA base	0.5360	0.3783
1–5	TransFR [38]	0.3711	0.1562
	MoE-2	0.3915	0.1662
	FedPASA base	0.3439	0.2045
6–20	TransFR [38]	0.3465	0.1395
	MoE-2	0.3541	0.1438
	FedPASA base	0.2631	0.0990
>20	TransFR [38]	0.3455	0.1361
	MoE-2	0.3544	0.1404
	FedPASA base	0.4956	0.2258

The unseen slice contains 20 nonempty evaluations (mean support: 13 users); other slices contain 30. Evaluations with zero hits are included.

E Hyperparameters and Robustness Analysis

Table 21: Validation-based parameter selection on Music→Movies (mean sampled HR@20 over three runs; eight rounds, 30 local steps). Stage 1 fixes $\lambda = 0.4, \rho = 0.2$; Stage 2 fixes the selected $\alpha = 0.85, \kappa = 0.5$. The final choice is $(0.85, 0.5, 0.2, 0.3)$.

$\alpha\backslash\kappa$	0.5	0.7	1.0
0.4	0.2807	0.2640	0.2517
0.55	0.3261	0.2871	0.2626
0.7	0.3657	0.3139	0.2721
0.85	0.3851	0.3541	0.2883

$\lambda\backslash\rho$	0.1	0.2	0.3
0.2	0.3917	0.3917	0.3925
0.4	0.3879	0.3851	0.3893
0.6	0.3842	0.3874	0.3817

E.1 Configuration

Defaults are DistilBERT-base-uncased ($d = 768$), two residual blocks (bottleneck 64, dropout 0.1), AdamW (learning rate 10^{-3}, weight decay 10^{-4}), and BPR batches of 32 positives with three negatives each (logit scale 10). Five clients use 12×60 rounds/local steps, or 8×30 for target Books. Adaptation uses $(\alpha, t, \kappa, \lambda) = (0.85, 0, 0.5, 0.2)$ and $(\rho, \tau_l, \tau_h) = (0.3, 1, 1.5)$. Proximal coefficients are 0.1 for FedProx, 0.01 for L2-SP, and 0 for FedPASA base. FedPASA and Prox-FT use 40 personalization steps with $\mu_p = 0.15$. FedMF uses 64-dimensional factors and 25×80 rounds/local steps.

Table 22: HR@20/NDCG@20 for Figure 5 (Music→Movies, three runs). (d): selected default.

Factor	Value	HR@20	NDCG@20
α	0.55	0.4543	0.1935
	0.7	0.4782	0.2079
	0.85	0.4938	0.2194 (d)
	1.0	0.4963	0.2223
κ	0.35	0.4998	0.2233
	0.5	0.4938	0.2194 (d)
	0.7	0.4842	0.2100
	1.0	0.4249	0.1779
λ	0.0	0.4563	0.2010
	0.1	0.4845	0.2110
	0.2	0.4938	0.2194 (d)
	0.4	0.4955	0.2208
ρ	0.0	0.4949	0.2190
	0.1	0.4952	0.2208
	0.2	0.4943	0.2199
	0.3	0.4938	0.2194 (d)
	0.5	0.4919	0.2208

E.2 Baseline Configuration

FFMSR uses BERT features and validation-based model selection. Pretraining and fine-tuning last 5 and 20 epochs for M–V, or 1 and 5 epochs for pairs involving Books. FedDCSR uses 10 communication rounds with three local epochs per round. The sequential benchmarks contain 10,634 Music, 89,254 Books, and 18,643 Movies items. Table 13 reports their full-ranking results.

Table 23: Robustness: 99-negative ranking, three-run FedPASA base means. Client and partition rows use Music→Movies; noise rows average six transfers.

Setting	HR@20	NDCG@20
5 clients	0.4940	0.2183
20 clients	0.5769	0.2788
50 clients	0.6035	0.2935
Activity split 0.1	0.4523	0.2059
Activity split 0.5	0.4839	0.2167
Popularity split 0.5	0.4895	0.2199
Fixed compute: 5 clients	0.2803	0.1089
Fixed compute: 20 clients	0.2460	0.0919
Fixed compute: 50 clients	0.2419	0.0889
Participation 0.2	0.2583	0.1034
Participation 0.5	0.2692	0.1074
Noise 0.1	0.4346	0.1938
Noise 0.2	0.4351	0.1944
Noise 0.3	0.4343	0.1941
LLaMA M→V	0.5091	0.2243
LLaMA B→V	0.4869	0.2203
LLaMA M→B	0.2519	0.0998

Table 24: Anchor granularity (99-negative ranking; five-run cross-domain means). The catalog gate avoids $|\mathcal{I}|$ trainable item logits and their federated updates, with accuracy close to per-item anchoring.

Anchoring	HR@20	NDCG@20
None	0.4346	0.1934
Catalog gate	0.4360	0.1944
Per-item gate	0.4365	0.1950

Table 25: Cosine distance from the frozen prior. Reference: TransFR.

Slice	Reference	No anchor	Base
Cross all	0.8916	0.9450	0.9841
Cross cold	0.8983	0.9721	0.9645

Fixed-compute and participation runs use eight rounds with $\alpha = 0.55$, $\kappa = 0.7$, $\lambda = 0.4$, and $\rho = 0.2$. The aggregate local steps per round are 150/160/150 for 5/20/50 clients. Participation sweeps use five clients; other rows use the selected configuration.